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REVIEW ARTICLE

Obstetrics

Sedentary behavior during pregnancy and risk of pre-eclampsia and/or gestational hypertension: A systematic review

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Abstract

Background: Despite improvements in perinatal care, pre-eclampsia remains a major cause of maternal and neonatal morbidity and mortality. Gestational hypertension and blood pressure are important intermediate endpoints for pre-eclampsia. Sedentary behavior has emerged as an independent risk factor for chronic disease, regardless of physical activity levels.

Objectives: To systematically review the evidence on sedentary behavior during pregnancy and in the year before conception and its effect on risk of pre-eclampsia, gestational hypertension, combined hypertensive disorders of pregnancy, and blood pressure.

Search Strategy: A systematic search of Ovid MEDLINE, Embase, Cochrane Central Register of Controlled Trials, Scopus, Web of Science, and EuropePMC preprints was performed up to May 25, 2023.

Selection Criteria: Observational and experimental studies investigating the association between sedentary behavior and pre-eclampsia, gestational hypertension, combined hypertensive disorders of pregnancy, and blood pressure in pregnant individuals.

Data Collection and Analysis: Effect estimates, a measure of uncertainty, and study and participant characteristics were extracted from study reports. A narrative synthesis is presented with forest plot visualizations. Quality assessment was conducted using the Newcastle-Ottawa Scale for cohort/case–control studies and the Appraisal Tool for Cross-Sectional Studies.

Main Results: Of the 12 observational studies meeting eligibility criteria, two studies reported on the outcome of gestational hypertension; no significant association was found between amount of time spent sedentary and gestational hypertension risk. Four studies reported on pre-eclampsia risk and found no association between sedentary behavior and pre-eclampsia risk. Four studies assessed a combined outcome of total hypertensive disorders of pregnancy; of these, one study reported greater odds of hypertensive disorders among sedentary individuals compared with non-sedentary individuals, and one study found that cases spent more time on the computer relative to controls. Five studies investigated blood pressure in pregnancy; of these, one

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study reported a positive correlation between systolic blood pressure and sedentary behavior, and one study reported higher second-trimester diastolic blood pressure in groups with high versus low sedentary behavior.

Conclusions: We found little evidence of an association between sedentary behavior and pre-eclampsia, gestational hypertension, or blood pressure. This is in contrast to the large body of evidence supporting the protective effect of physical activity on hypertensive disorders of pregnancy. Heterogeneity in the methods used to estimate sedentary behavior limited the ability to quantify this association. Additional high-quality observational studies or randomized trials are warranted to reach more definitive conclusions.

KEYWORDS

blood pressure, gestational hypertension, pre-eclampsia, pregnancy, sedentary behavior, systematic review

1 | INTRODUCTION

Pre-eclampsia (PE) is a severe complication of pregnancy, which involves a new onset of hypertension and proteinuria after 20 weeks' gestation, with or without other maternal organ dysfunction.[1] Despite improvements in perinatal care, the prevalence of pre-eclampsia has improved little, affecting an estimated 2%–10% of pregnancies worldwide and remaining a leading cause of maternal and neonatal morbidity and mortality.[2] Specifically, pre-eclampsia is a major risk factor for fetal growth restriction, placental abruption, preterm birth, pulmonary edema, liver failure, eclampsia, and cardiovascular events, among other severe complications.[1] An estimated 15% of maternal deaths can be attributed to pre-eclampsia, with the majority occurring in low- and middle-income countries.[1,3,4] Gestational hypertension and blood pressure are important intermediate endpoints, with up to 50% of individuals with gestational hypertension eventually developing proteinuria and other end-organ dysfunction consistent with the diagnosis of pre-eclampsia.[5]

At present, there are no reliable biomarkers, preventative strategies for disease progression, or treatments for pre-eclampsia other than delivery.[1] A 2016 systematic review identified a number of risk factors for pre-eclampsia, including prior pre-eclampsia (eight-fold increase in risk; relative risk [RR] [95% confidence interval, CI] = 8.4 [7.1–9.9]), chronic hypertension (five-fold increase in risk; RR [95% CI] = 5.1 [4.0–6.5]), pre-gestational diabetes (four-fold increase in risk; RR [95% CI] = 3.7 [3.1–4.3]), multi-fetal pregnancy (three-fold increase in risk; RR [95% CI] = 2.9 [2.6–3.1]), pre-pregnancy body mass index > 30 (three-fold increase in risk; RR [95% CI] = 2.8 [2.6–3.1]), and others.[6] Other studies have identified risk factors for hypertensive disorders of pregnancy to include obesity, family history of hypertension, alcohol intake, heart failure, stroke and left ventricular hypertrophy, and smoking.[7–10]

Physical activity is associated with both weight loss and weight maintenance[11] and is protective against hypertension, type 2 diabetes, and cardiovascular disease.[12] It is hypothesized that physical activity may result in reduced blood pressure levels and improved cardiovascular conditioning in pregnant individuals. Regarding pre-eclampsia, physical activity may exert a protective effect by reducing oxidative stress, which stimulates vascularity and placental growth, while preventing endothelial dysfunction.[13,14] Many studies and systematic reviews have been published in the past two decades regarding the association between physical activity and the risk of pre-eclampsia and/or gestational hypertension, demonstrating a protective effect of physical activity against risk of gestational hypertensive disorders.[15,16]

Conversely, substantially less research has been conducted on the association between sedentary behavior and hypertensive disorders of pregnancy. Sedentary behavior is defined as activities with an energy expenditure below 1.5 metabolic equivalent units (i.e., activities that expend very low energy), such as sitting, driving a car, or watching television.[17] In the general adult population, approximately 55%–60% of awake time is spent in sedentary behaviors.[18–20] This is similar among pregnant individuals who spend approximately 57%–65% of awake time sedentary.[21–23] Sedentary behavior has emerged as an important and independent risk factor for chronic disease and is linked to all-cause mortality, cardiovascular disease, and type 2 diabetes in adults, regardless of physical activity levels.[24] Excessive sedentary behavior is still common among those who reach the recommended daily levels of physical activity.[25] It is for this reason that the association between sedentary behavior specifically—not just low levels of physical activity—and pre-eclampsia should be investigated, although physical inactivity is occasionally mistakenly used as a marker for sedentary behavior.

2 | OBJECTIVE

In this study, we systematically reviewed the available evidence on time spent engaging in sedentary behavior during pregnancy, and

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in the year immediately prior to conception, and its effect on subsequent risk of pre-eclampsia, risk of gestational hypertension, and blood pressure.

3 | METHODS

This systematic review is reported in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) reporting guideline. The protocol was registered in PROSPERO (CRD42023427233).

3.1 | Search strategy

Electronic database searches

The search was conducted in Ovid MEDLINE, Ovid Embase, Ovid Cochrane Central Register of Controlled Trials (CENTRAL), Scopus, and Web of Science (Science Citation Index Expanded; Social Sciences Citation Index) databases from their inception to May 25, 2023. Searches included a combination of subject headings (e.g., medical subject headings (MeSH) in MEDLINE) and free-text keywords (.mp). Detailed search strategies are provided in Appendix S1. The search was based on the outcome of interest—pre-eclampsia, including gestational hypertension and blood pressure as intermediate endpoints—and the exposure of interest: sedentary behavior, including screen time, sitting, and inactivity. Searches were not restricted by study design, publication date, language or format, age of participants, or geographic location of study conduct. Non-English language articles were translated by a native speaker of that language to determine potential eligibility.

The gray literature was searched to identify unpublished study reports—specifically, EuropePMC for preprints. This search was restricted to the previous 2 years in order to avoid duplication with older results that may have already been published.

Reference lists

Reference lists of all included studies, as well as any previous systematic reviews on this topic, were checked by hand to identify any additional studies that may have been missed by the electronic database searches.

3.2 | Eligibility criteria

Studies were considered eligible for inclusion if the population comprised pregnant individuals, with or without comorbidities. There were no restrictions in terms of age, setting (i.e., community vs. clinical), geographical location, or ethnicity. Inclusion of studies was restricted to those with biologically female participants, with no restrictions on gender. Only human studies were included.

The exposure was time spent engaging in sedentary behavior during pregnancy or in the year prior to conception. If a study interval extended to earlier than 1 year prior to conception, the study was included if the majority of the time interval overlapped with our desired exposure window from 1 year prior to conception to during pregnancy. For studies reporting sedentary time on a categorical scale, "high" levels were taken to be the highest reported category of sedentary time and "low" levels were taken to be the lowest reported category of sedentary time in each study. If sedentary time was reported on a continuous scale, the exposure of time spent sedentary was summarized and analyzed on a continuum. Sedentary behavior was either self-reported through questionnaire or recall methods (i.e., time spent sitting, watching TV, on the computer) or objectively measured through inclinometers or accelerometers. Studies that only reported physical inactivity as a proxy for sedentariness, rather than a measure of time spent sedentary, were excluded.

The outcome was a pre-eclampsia diagnosis, using guidelines such as those from the American College of Obstetricians and Gynecologists (ACOG) or the National Heart, Lung, and Blood Institute.[5,26] Use of ICD codes and diagnosis from a physician were also accepted. ACOG defines pre-eclampsia as a disorder of pregnancy associated with new-onset hypertension, occurring after 20 weeks' gestation, often accompanied by new-onset proteinuria; however, hypertension and other signs or symptoms of pre-eclampsia may present in the absence of proteinuria.[5] We also included gestational hypertension, defined as a systolic blood pressure ≥ 140 mmHg and/or diastolic blood pressure ≥ 90 mmHg, measured on two occasions at least 4 hours apart after 20 weeks' gestation with previously normal blood pressure.[5] Both pre-eclampsia and gestational hypertension are conditions that fall under the broad class of hypertensive disorders of pregnancy, and clinical management is typically the same. Additional outcomes of systolic and diastolic blood pressure, measured on a continuous scale, were also included in this review.

Primarily original case–control (nested or not) and cohort observational study designs with an exposed group and a comparator group were included. Cross-sectional designs were considered if an assessment of sedentary time during our desired exposure window was available. Due to the nature of the outcome of interest and its implications for morbidity and mortality, included studies were primarily observational and we did not find randomized controlled trials or experimental evidence. Review articles, case reports, and case series study designs were excluded.

3.3 | Study selection

All citations were imported into EndNote and deduplicated. Citations were then uploaded to Covidence to facilitate screening.[27] Titles and abstracts were screened independently by two reviewers (LXP and SM) against eligibility criteria. Full-texts of all possibly eligible studies were screened independently by the same two

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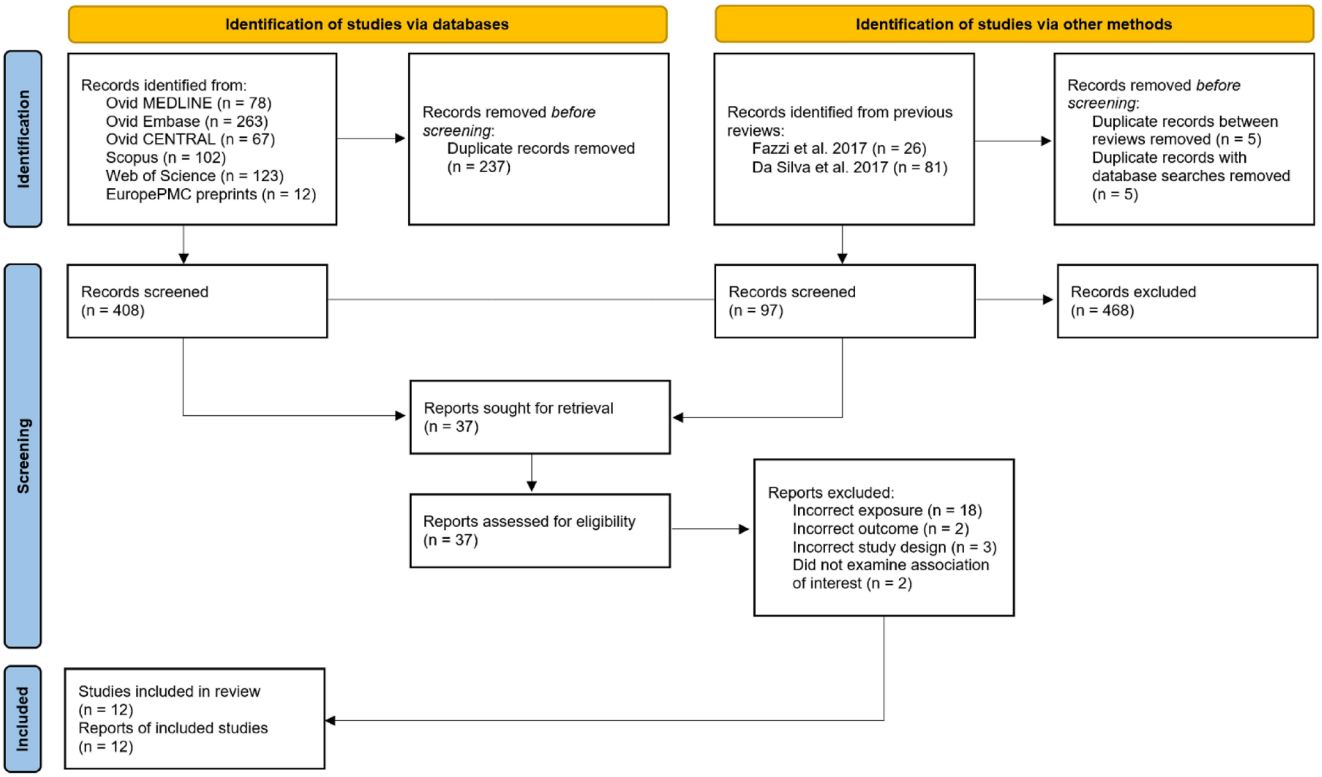

FIGURE 1 PRISMA flow diagram of study identification and selection.

reviewers (LXP and SM) against eligibility criteria. Online supplementary materials were consulted if the information provided in the main published article was insufficient to assess eligibility. Studies that did not meet eligibility criteria were excluded, with reasons for exclusion recorded. In the case of multiple publications on the same cohort, only the study with the longest follow-up period was included. During both title-abstract and full-text screening stages, any disagreements were resolved by discussion between the two reviewers to reach consensus. The screening and selection process is documented in the PRISMA flow diagram in Figure 1.

3.4 | Data extraction

Information was extracted from the full texts of all selected studies using a prespecified Microsoft Excel data extraction form designed by the review team. Extracted data included general information, study characteristics, participant characteristics, exposure details, outcome details, statistical analyses, and reported effect measures. Data extraction was conducted by two independent reviewers (LXP and SM), with discrepancies resolved by discussion to reach consensus.

In the case of studies reporting multiple analyses with either the exposure or the outcome measured in more than one manner, objective data were prioritized over self-reported data; however, information on all potentially relevant comparisons was extracted. Study authors were contacted if key information relevant to the present review was missing or unclear from the published reports of included studies.

Effect estimates were extracted directly from study reports, along with a measure of uncertainty (e.g., standard error, 95% confidence interval [CI], *P*-value). Effect estimates included *t*-values, *z*-scores, beta coefficients, correlation coefficients, and odds ratios (ORs). If such information was not included, any other relevant information reported by study authors in study reports was extracted; no post-hoc calculations were performed by the review team.

3.5 | Data synthesis

Meta-analysis was unable to be conducted due to an insufficient number of included studies assessing the same outcome measure and using similar exposure definitions. Therefore, for each outcome measure of interest, a narrative synthesis of results was presented. For primary outcome measures of gestational hypertension and preeclampsia, effect measures were displayed using forest plots, without a pooled summary estimate. Forest plots were generated using R version 4.1.2 (R Core Team 2021).

3.6 | Study quality assessment

The Newcastle-Ottawa Scale was used to assess the quality of included cohort and case–control studies,[28] and the Appraisal Tool for Cross-Sectional Studies (AXIS) was used for cross-sectional studies.[29] Two reviewers (LXP and SM) independently assessed the

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quality of included studies with disagreements resolved through discussion to reach consensus.

4 | RESULTS

We identified 408 studies through electronic searches. From our manual search of two systematic reviews, we screened an additional 97 studies. Of these, 468 were excluded on the basis of title and abstract, leaving 37 studies for further assessment. We excluded 25 studies following full-text assessment: 18 had an incorrect exposure, two had an incorrect outcome, three were an incorrect study design, and two did not examine the association of interest (see Appendix S2 for further details on reasons for exclusion). Twelve studies met the inclusion criteria and were included in the analysis.[30–41] Table 1 provides a summary of the included studies. Regarding study design, four studies were case–control,[35,38–40] five were cohort,[31–33,37,41] and three were cross-sectional.[30,34,36] Seven countries were represented by the studies. The studies were published between 2004 and 2022, all in English. Table 2 summarizes extracted effect estimates from included studies along with covariates included in adjusted analyses and an assessment of study quality.

4.1 | Gestational hypertension

The association between sedentary behavior and risk of gestational hypertension was reported in two studies. Saftlas et al. used two assessments of sedentary behavior self-reported through an interview: (1) number of hours per day spent sitting; and (2) work activity levels derived by calculating the percentage of each participant's work day spent sitting.[39] No significant association was found between amount of time spent sitting and gestational hypertension risk ($P=0.73$; Figure 2). Spracklen et al. also assessed two self-reported measures of sedentary behavior: (1) amount of time per day spent sitting at work (continuous and categorized); and (2) a composite sitting variable approximating the total time spent sedentary per day, calculated using self-reported number of hours spent on feet at work and number of hours spent on feet away from work.[40] Similarly, no significant association was found between sedentary behavior and risk of gestational hypertension (Figure 2). Chasan-Taber et al. also investigated the outcome of gestational hypertension; however, due to low incidence ($n=17$, 1.4%), authors were unable to evaluate this outcome separately from other hypertensive disorders of pregnancy.[31]

4.2 | Pre-eclampsia

The association between sedentary behavior and risk of pre-eclampsia was investigated in four studies. In addition to the outcome of gestational hypertension, Saftlas et al. and Spracklen et al. also investigated pre-eclampsia using the same assessments of sedentary behavior as described earlier; however, no significant associations were found (Figure 2).[39,40] Chasan-Taber et al. examined the association between sedentary behavior during pre- and early pregnancy self-reported on the Pregnancy Physical Activity Questionnaire (PPAQ) and risk of developing pre-eclampsia among Hispanic women and found no significant difference in risk between sedentary behavior tertiles (Figure 2).[31] Do et al. also assessed sedentary behavior through self-report on the PPAQ and found a significant association between weekly energy expenditure (MET-h/week) in sedentary activities and pre-eclampsia risk (unadjusted OR [95% CI] 1.04 [1.00–1.08], $P=0.03$); however, after adjusting for parity, smoking and diastolic blood pressure, the association was no longer significant (adjusted OR [95% CI] 1.03 [0.99–1.08], $P=0.13$).[32]

4.3 | Total hypertensive disorders of pregnancy

Four studies reported on the association between sedentary behavior and total hypertensive disorders of pregnancy. Three studies[31,34,37] defined total hypertensive disorders of pregnancy as combined pre-eclampsia and gestational hypertension; however, Liu et al. defined it as combined pre-eclampsia, gestational hypertension, eclampsia, and either pre-eclampsia or gestational hypertension in addition to chronic hypertension. Meander et al. assessed sedentary behavior as self-reported time spent sitting down and reported two analyses, continuous and binary (≥7 vs. <7 h).[37] The continuous exposure was borderline statistically significant (unadjusted OR [95% CI] 1.21 [1.00–1.50]), whereas the binary exposure was statistically significant (unadjusted OR [95% CI] 1.65 [1.03–2.62]). Chasan-Taber et al. reported effect estimates for sedentary behavior in pre-pregnancy and early pregnancy, separately, although neither was significantly associated with total hypertensive disorders of pregnancy.[31] Similarly, Lawan et al. assessed sedentary time through self-report on the PPAQ and found that sedentary behavior did not differ significantly between participants with pre-eclampsia or gestational hypertension relative to those without ($P=0.162$).[34] Liu et al. defined sedentary behavior as self-reported time spent using a computer and time spent using a smartphone and/or tablet computer.[35] Comparing computer time between cases and controls, they found that cases spent more time on the computer relative to controls (median [interquartile range, IQR] 0 [0–6.0] vs. 3.0 [0–7.5], $P=0.003$). However, time spent on smartphone and/or tablet computers was not significantly different between cases and controls (median [IQR] 5.50 [3.50–7.88] vs. 5.50 [3.50–7.50], $P=0.089$).

4.4 | Blood pressure

The association of sedentary behavior with blood pressure in pregnancy was investigated in five studies. All included studies reported this association for blood pressure measured at one point in time, except for Lane et al., who reported this association for blood pressure measured in each trimester,[33] although we only discuss reports for

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TABLE 1 Characteristics of studies included in the systematic review.

First author, year	Country	Age (years)	Sample size	Exposure definition	Exposure measurement	Outcome definition	Outcome measurement	Statistical methods
Case–control studies								
Liu, 2021[35]	China	Median: 30	357	Time per day on computer (h/day) Time per day on smartphone and/or tablet (h/day)	Interview	GH, PE, eclampsia, PE with chronic hypertension, and GH with chronic hypertension: categorized together	Medical records and/or questionnaire	Mann–Whitney *U*-test
Raguema, 2020[38]	Tunisia	Mean: 30.7	106	Sedentary activities lower than 2.0 METs (MET-h/week)	PPAQ	Systolic blood pressure (mmHg) Diastolic blood pressure (mmHg) PE: hypertension (systolic blood pressure ≥ 140 mmHg and diastolic blood pressure ≥ 90 mmHg on two separate occasions, at least 4 h apart) and proteinuria (≥1+ by dipstick or ≥300 mg/24 h) after 20 weeks of pregnancy	Medical records	Pearson correlation test Mann–Whitney *U*-test
Saftlas, 2004[39]	USA	PE mean: 30.8 GH mean: 30.5 Control mean: 31.4	2638	Proportional sitting: fraction of work day spent sitting Time per day spent sitting (h/day) Proportional sitting: fraction of work day spent sitting Time per day spent sitting (h/day)	Interview	PE: GH with proteinuria (two or more dipstick readings of 2+ or greater, one catheter sample reading of 2+ or greater, or a 24-h urine collection containing at least 300 mg of protein) GH: systolic blood pressure of at least 140 mmHg or diastolic blood pressure of at least 90 mmHg on at least two occasions taken at least 6 hours apart after 20 weeks' gestation	Medical records	Univariate and multivariable logistic regression *t*-test Univariate and multivariable logistic regression *t*-test
Spracklen, 2016[40]	USA	Mean: 26.4	673	Time per day sitting at work (h/day), continuous and categorical Composite variable of time per day sitting (h/day), continuous and categorical Time per day sitting at work (h/day), continuous and categorical Composite variable of time per day sitting (h/day), continuous and categorical	Computer-assisted telephone interview	PE: sustained de novo hypertension (≥140 mmHg systolic or ≥90 mmHg diastolic on two or more occasions at least 6 h apart) with onset after the 20th week of gestation, and proteinuria, defined as urinary protein concentrations ≥30 mg/dL (i.e., dipstick value of 1+ from two or more specimens collected ≥4 h apart; or at least one ≥1 dipstick value of 2+ or greater; or a catheter dipstick value ≥1+; or a 24-h urine collection ≥300 mg of protein) GH: sustained de novo hypertension after 20 weeks gestation with no evidence of proteinuria	Medical records	Univariate and multivariable multinomial logistic regression

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TABLE 1 (Continued)

First author, year	Country	Age (years)	Sample size	Exposure definition	Exposure measurement	Outcome definition	Outcome measurement	Statistical methods
Cohort studies								
Chasan-Taber, 2015[31]	USA	31.5% <19 years; 39.0% 19–23 years; 17.8% 24–29 years; 11.7% ≥30 years	1212 in pre-pregnancy analyses and 904 in early pregnancy analyses	MET-h/day watching TV/videos or sitting/standing at home in pre-pregnancy MET-h/day watching TV/videos or sitting/standing at home in early pregnancy MET-h/day watching TV/videos or sitting/standing at home in pre-pregnancy MET-h/day watching TV/videos or sitting/standing at home in early pregnancy	PPAQ	PE: blood pressure greater than 140/90 mmHg on two occasions after 20 weeks gestation in a previously normotensive woman, with proteinuria or symptoms consistent with PE, other abnormal laboratory values such as low platelets or elevated liver function tests or the presence of eclampsia PE and GH: categorized together; GH defined as two blood pressure measurements >140/90 mmHg after 20 weeks gestation in a previously normotensive woman, with no laboratory evidence or symptoms of PE; PE defined as blood pressure> 140/90 mmHg on two occasions after 20 weeks gestation in a previously normotensive woman, with proteinuria or symptoms consistent with PE, other abnormal laboratory values such as low platelets or elevated liver function tests or the presence of eclampsia	Medical records, with diagnoses confirmed by an obstetrician	Univariate, age-adjusted, and multivariable logistic regression
Do, 2020[32]	Denmark	PE mean: 31 Control mean: 32	189	Inactivity/sedentary behavior (<1.5 METs) including watching TV/videos and sitting at home reading, talking or on the phone (MET-h/week)	PPAQ	PE: blood pressure ≥ 140 mmHg systolic and/or 90 mmHg diastolic measured twice at least 4 h apart after 20 gestational weeks and proteinuria ≥1+ on urine dipstick and/or new onset of symptoms of organ dysfunction	Medical records	Univariate and multivariable logistic regression
Lane, 2022[33]	USA	Low SED mean: 33 Medium SED mean: 32 High SED mean: 30	111	Trajectory groups of time spent in a sitting or reclining position that requires little energy expenditure	Activity monitor	Systolic blood pressure in second trimester, measured at study visit Diastolic blood pressure in second trimester, measured at study visit Systolic blood pressure in second trimester, measured at clinic Diastolic blood pressure in second semester, measured at clinic Systolic blood pressure in third trimester, measured at study visit Diastolic blood pressure in third trimester, measured at study visit Systolic blood pressure in third trimester, measured at clinic Diastolic blood pressure in third trimester, measured at clinic	Blood pressure monitor	Multivariable linear regression

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TABLE 1 (Continued)

First author, year	Country	Age (years)	Sample size	Exposure definition	Exposure measurement	Outcome definition	Outcome measurement	Statistical methods
Meander, 2021[37]	Sweden	Lowest quartile of PA mean: 31.0 Highest quartile of PA mean: 30.9	2201	Time spent sitting down, not including sleeping (h/day), continuous and dichotomized	Validated question	PE and GH: categorized together; any of the following ICD10 codes O14.0, O14.1, O14.1A, O14.1B, O14.1X, O14.2, O14.9, O13.9	Medical records	Univariate and multivariable logistic regression
Wallace, 2022[41]	USA	Mean: 30.85	131	Time spent sitting per day (h/day)	SB2D	Systolic blood pressure readings (mmHg) Diastolic blood pressure readings (mmHg)	Blood pressure monitor	Multivariable linear regression
Cross-sectional studies								
Aburezq, 2020[30]	Kuwait	Mean: 30.13	653	Time spent in sedentary activities (MET-h/week)	PPAQ	Systolic blood pressure Diastolic blood pressure	Self-reported on questionnaire	Spearman correlation test
Lawan, 2020[34]	Nigeria	Median: 26	130	Sedentary time < 1.5 METs	PPAQ	GH and PE: categorized together; systolic blood pressure of ≥140 mmHg or a diastolic blood pressure of ≥90 mmHg on two or more occasions, taken at least 6 h apart after 20 weeks of gestation or the presence of GH with proteinuria (24-h urine collection containing at least 300 mg of protein or the excretion of ≥300 mg/L of concentration of protein (1+ on the dipstick), in two or more samples taken randomly (4–6 h apart)	Medical records	*t*-test
Loprinzi, 2013[36]	USA	Mean: 28.4	206	Time spent in sedentary behavior (min/day)	Accelerometer	Systolic blood pressure (mmHg) Diastolic blood pressure (mmHg)	Blood pressure monitor	Multivariable linear regression

Abbreviations: GH, gestational hypertension; MET, metabolic equivalent of task; PA, physical activity; PE, pre-eclampsia; PPAQ, Pregnancy Physical Activity Questionnaire; SB2D, Sedentary Behavior Two Domain Questionnaire; SED, sedentary behavior.

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TABLE 2 Extracted effect estimates and statistical analyses from included studies along with quality assessment summary.

First author, year	Exposure	Outcome	Effect estimate	Effect size from unadjusted analyses	Effect size from maximally adjusted analyses	Confounders included in analyses	Quality assessment[a]
Case–control studies							
Liu, 2021[35]	Time per day on computer (h/day)	GH, PE, eclampsia, PE with chronic hypertension, and GH with chronic hypertension: categorized together	*Z*-score	−3.00 ($P=0.003$)	N/A	N/A	5 stars: 4 for selection; 0 for comparability; 1 for exposure
	Time per day on smartphone and/or tablet (h/day)		*Z*-score	−1.67 ($P=0.089$)	N/A	N/A	5 stars: 4 for selection; 0 for comparability; 1 for exposure
Raguema, 2020[38]	Sedentary activities lower than 2.0 METs (MET-h/week)	PE	Mean (SD)	44.05 (28.48) in PE cases 28.06 (20.64) in controls ($P<0.001$)	N/A	N/A	6 stars: 4 for selection; 1 for comparability; 1 for exposure
		Systolic blood pressure	Pearson correlation coefficient	0.25 ($0.001<P<0.05$)	N/A	N/A	6 stars: 4 for selection; 1 for comparability; 1 for exposure
		Diastolic blood pressure	Pearson correlation coefficient	0.08 ($P>0.05$)	N/A	N/A	6 stars: 4 for selection; 1 for comparability; 1 for exposure
Saftlas, 2004[39]	Time per day spent sitting (h/day)	PE	Mean (SD)	4.3 (2.5) in PE group; 4.0 (2.4) in normotensive group ($P=0.41$)	N/A	N/A	5 stars: 4 for selection; 0 for comparability; 1 for exposure
		GH	Mean (SD)	3.9 (2.3) in GH group; 4.0 (2.4) in normotensive group ($P=0.73$)	N/A	N/A	5 stars: 4 for selection; 0 for comparability; 1 for exposure
	Proportional sitting: fraction of work day spent sitting	PE	Odds ratio	Least sitting vs. highest sitting: 0.68	0.72 (95% CI 0.32–1.59)	LTPA during pregnancy, BMI, education, parity, abortion history	6 stars: 4 for selection; 1 for comparability; 1 for exposure
		GH	Odds ratio	Least sitting vs. highest sitting: 1.21	1.26 (95% CI 0.83–1.91)	LTPA during pregnancy, age at delivery, BMI, education, parity, abortion history	7 stars: 4 for selection; 2 for comparability; 1 for exposure

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TABLE 2 (Continued)

First author, year	Exposure	Outcome	Effect estimate	Effect size from unadjusted analyses	Effect size from maximally adjusted analyses	Confounders included in analyses	Quality assessment[a]
Spracklen, 2016[40]	Time per day sitting at work (h/day), continuous	PE	Odds ratio	1.05 (95% CI 0.98–1.13)	1.06 (95% CI 0.98–1.13)	BMI	6 stars: 4 for selection; 1 for comparability; 1 for exposure
		GH	Odds ratio	1.05 (95% CI 0.98–1.12)	1.05 (95% CI 0.97–1.12)	BMI	6 stars: 4 for selection; 1 for comparability; 1 for exposure
	Time per day sitting at work (h/day), categorical	PE	Odds ratio	1.46 (95% CI 0.80–2.65)	1.52 (95% CI 0.81–2.85)	BMI	6 stars: 4 for selection; 1 for comparability; 1 for exposure
		GH	Odds ratio	1.31 (95% CI 0.72–2.36)	1.28 (95% CI 0.69–2.37)	BMI	6 stars: 4 for selection; 1 for comparability; 1 for exposure
	Composite variable of time per day sitting (h/day), continuous	PE	Odds ratio	1.04 (95% CI 0.99–1.10)	1.03 (95% CI 0.99–1.07)	BMI	6 stars: 4 for selection; 1 for comparability; 1 for exposure
		GH	Odds ratio	1.00 (95% CI 0.95–1.05)	1.02 (95% CI 0.98–1.06)	BMI	6 stars: 4 for selection; 1 for comparability; 1 for exposure
	Composite variable of time per day sitting (h/day), categorical	PE: sustained de novo hypertension (≥140 mmHg systolic or ≥ 90 mmHg diastolic on two or more occasions at least 6 h apart) with onset after the 20th week of gestation, and proteinuria, defined as urinary protein concentrations ≥30 mg/dL (i.e., dipstick value of 1+ from two or more specimens collected ≥4 h apart; or at least one ≥1 dipstick value of 2+ or greater; or a catheter dipstick value ≥1+; or a 24-h urine collection ≥300 mg of protein)	Odds ratio	1.40 (95% CI 0.86–2.29)	1.38 (95% CI 0.83–2.30)	BMI	6 stars: 4 for selection; 1 for comparability; 1 for exposure
		GH: sustained de novo hypertension after 20 weeks gestation with no evidence of proteinuria	Odds ratio	1.01 (95% CI 0.62–1.65)	1.01 (95% CI 0.61–1.67)	BMI	6 stars: 4 for selection; 1 for comparability; 1 for exposure

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TABLE 2 (Continued)

First author, year	Exposure	Outcome	Effect estimate	Effect size from unadjusted analyses	Effect size from maximally adjusted analyses	Confounders included in analyses	Quality assessment[a]
Cohort studies							
Chasan-Taber, 2015[31]	MET-h/day watching TV/videos or sitting/standing at home in pre-pregnancy	GH	Due to low incidence of GH ($n = 17$, 1.4%), unable to evaluate outcome separately				
		PE	Odds ratio	No significant associations between total pre–pregnancy physical activity, or intensity or type of activity, and risk of hypertensive disorders	3rd tertile vs. 1st tertile: 1.5 (95% CI 0.6–3.6)	Age, pre-pregnancy BMI, parity	9 stars: 4 for selection; 2 for comparability; 3 for outcome
		PE and GH: categorized together	Odds ratio	No significant associations between total pre-pregnancy physical activity, or intensity or type of activity, and risk of hypertensive disorders	3rd tertile vs. 1st tertile: 1.4 (95% CI 0.7–2.9)	Age, pre-pregnancy BMI, parity	9 stars: 4 for selection; 2 for comparability; 3 for outcome
	MET-h/day watching TV/videos or sitting/standing at home in early pregnancy	GH	Due to low incidence of GH ($n = 17$, 1.4%), unable to evaluate outcome separately				
		PE	Odds ratio	No significant associations between total pre-pregnancy physical activity, or intensity or type of activity, and risk of hypertensive disorders	3rd tertile vs. 1st tertile: 1.0 (95% CI 0.3–3.0)	Age, pre-pregnancy BMI, parity	9 stars: 4 for selection; 2 for comparability; 3 for outcome
		PE and GH: categorized together	Odds ratio	No significant associations between total pre-pregnancy physical activity, or intensity or type of activity, and risk of hypertensive disorders	3rd tertile vs. 1st tertile: 1.2 (95% CI 0.5–3.0)	Age, pre-pregnancy BMI, parity	9 stars: 4 for selection; 2 for comparability; 3 for outcome
Do, 2020[32]	Inactivity/sedentary behavior (<1.5 METs) including watching TV/videos and sitting at home reading, talking or on the phone (MET-h/week)	PE	Odds ratio	1.04 (95% CI 1.00–1.08)	1.03 (95% CI 0.99–1.08)	Parity, smoking, diastolic office blood pressure at 10 weeks gestation	6 stars: 3 for selection; 1 for comparability; 2 for outcome

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TABLE 2 (Continued)

First author, year	Exposure	Outcome	Effect estimate	Effect size from unadjusted analyses	Effect size from maximally adjusted analyses	Confounders included in analyses	Quality assessment[a]
Lane, 2022[33]	Trajectory groups of time spent in a sitting or reclining position that requires little energy expenditure	Study visit systolic blood pressure (second trimester)	Beta coefficient	N/A	High SED vs. low SED: 1.5 (95% CI −3.4–6.5)	Maternal age, pre-pregnancy BMI, black race, education, gestational age	7 stars: 3 for selection; 2 for comparability; 2 for outcome
		Study visit diastolic blood pressure (second trimester)	Beta coefficient	N/A	High SED vs. low SED: 4.9 (95% CI 1.6–8.2)	Maternal age, pre-pregnancy BMI, black race, education, gestational age	7 stars: 3 for selection; 2 for comparability; 2 for outcome
		Clinic systolic blood pressure (second trimester)	Beta coefficient	No difference between SED trajectory groups ($P > 0.09$)	High SED vs. low SED: 0.2 (95% CI −4.7–5.2)	Maternal age, pre-pregnancy BMI, black race, education, gestational age	7 stars: 3 for selection; 2 for comparability; 2 for outcome
		Clinic diastolic blood pressure (second trimester)	Beta coefficient	No difference between SED trajectory groups ($P > 0.09$)	High SED vs. low SED: 3.7 (95% CI −0.7–8.2)	Maternal age, pre-pregnancy BMI, black race, education, gestational age	7 stars: 3 for selection; 2 for comparability; 2 for outcome
		Study visit systolic blood pressure (third trimester)	Beta coefficient	N/A	High SED vs. low SED: 2.6 (95% CI −3.2–8.5)	Maternal age, pre-pregnancy BMI, black race, education, gestational age	7 stars: 3 for selection; 2 for comparability; 2 for outcome
		Study visit diastolic blood pressure (third trimester)	Beta coefficient	N/A	High SED vs. low SED: 2.9 (95% CI −1.4–7.2)	Maternal age, pre-pregnancy BMI, black race, education, gestational age	7 stars: 3 for selection; 2 for comparability; 2 for outcome
		Clinic systolic blood pressure (third trimester)	Beta coefficient	No difference between SED trajectory groups ($P > 0.09$)	High SED vs. low SED: −1.0 (95% CI −6.3–4.2)	Maternal age, pre-pregnancy BMI, black race, education, gestational age	7 stars: 3 for selection; 2 for comparability; 2 for outcome
		Clinic diastolic blood pressure (third trimester)	Beta coefficient	No difference between SED trajectory groups ($P > 0.09$)	High SED vs. low SED: 3.2 (95% CI −1.1–7.6)	Maternal age, pre-pregnancy BMI, black race, education, gestational age	7 stars: 3 for selection; 2 for comparability; 2 for outcome

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TABLE 2 (Continued)

First author, year	Exposure	Outcome	Effect estimate	Effect size from unadjusted analyses	Effect size from maximally adjusted analyses	Confounders included in analyses	Quality assessment[a]
Meander, 2021[37]	Time spent sitting down, not including sleeping (h/day), continuous	PE and pregnancy-induced hypertension: categorized together	Odds ratio	1.21 (95% CI 1.00–1.50)	N/A	N/A	4 stars: 2 for selection; 0 for comparability; 2 for outcome
	Time spent sitting down, not including sleeping (h/d), dichotomized (≥7 h/day vs. <7 h/day)	PE and pregnancy-induced hypertension: categorized together	Odds ratio	≥7 h/d vs. <7 h/d: 1.65 (95% CI 1.03–2.62)	After adjusting for BMI and parity, result was no longer significant	BMI, parity	5 stars: 2 for selection; 1 for comparability; 2 for outcome
Wallace, 2022[41]	Time spent sitting per day (h/day)	Systolic blood pressure (second trimester)	Standardized beta coefficient	N/A	0.030 ($P=0.74$)	Pre-pregnancy BMI, history of GH	6 stars: 3 for selection; 1 for comparability; 2 for outcome
		Diastolic blood pressure (second trimester)	Standardized beta coefficient	N/A	−0.050 ($P=0.55$)	Age, pre-pregnancy BMI, history of GH	7 stars: 3 for selection; 2 for comparability; 2 for outcome
		Systolic blood pressure (third trimester)	Standardized beta coefficient	N/A	0.075 ($P=0.45$)	Pre-pregnancy BMI, history of GH	6 stars: 3 for selection; 1 for comparability; 2 for outcome
		Diastolic blood pressure (third trimester)	Standardized beta coefficient	N/A	0.001 ($P=0.99$)	Age, pre-pregnancy BMI, history of GH	7 stars: 3 for selection; 2 for comparability; 2 for outcome
Cross-sectional studies							
Aburezq, 2020[30]	Time spent in sedentary activities (MET-h/week)	Systolic blood pressure	Spearman correlation coefficient	0.002 ($P=0.963$)	N/A	N/A	13/20 "yes"
		Diastolic blood pressure	Spearman correlation coefficient	−0.043 ($P=0.275$)	N/A	N/A	13/20 "yes"
Lawan, 2020[34]	Sedentary time<1.5 METs	Hypertension status: GH and PE categorized together	*t*-value	1.63 ($P=0.162$)	N/A	N/A	16/20 "yes"

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TABLE 2 (Continued)

First author, year	Exposure	Outcome	Effect estimate	Effect size from unadjusted analyses	Effect size from maximally adjusted analyses	Confounders included in analyses	Quality assessment[a]
Loprinzi, 2013[36]	Time spent in sedentary behavior (min/day)	Systolic blood pressure	Beta coefficient	N/A	−0.004 (95% CI −0.01–0.007)	Age, smoking, education, marital status, poverty to income ratio, race/ethnicity, BMI, gestational age, MVPA	16/20 "yes"
		Diastolic blood pressure	Beta coefficient	N/A	0.002 (95% CI −0.02–0.03)	Age, smoking, education, marital status, poverty to income ratio, race/ethnicity, BMI, gestational age, MVPA	16/20 "yes"

Abbreviations: BMI, body mass index (calculated as weight in kilograms divided by the square of height in meters); CI, confidence interval; GH, gestational hypertension; LTPA, leisure-time physical activity; MET, metabolic equivalent of task; MVPA, moderate-to-vigorous-intensity physical activity; PE, pre-eclampsia; SED, sedentary behavior.

[a]Study quality assessment performed using the Newcastle-Ottawa Scale for cohort and case–control studies and the Appraisal Tool for Cross-Sectional Studies (AXIS) for cross-sectional studies. For the Newcastle-Ottawa Scale, studies were required to control for both age and BMI to be awarded one star for the "comparability" category; studies were awarded an additional star if they controlled for any additional factor. Studies only controlling for one of the two were awarded only one star for comparability. Quality is summarized as the total number of stars awarded. For AXIS, quality is summarized as the total number of questions awarded a "yes" response. For detailed quality assessments for each included study, refer to Appendix S3.

the second and third trimesters here. Both Lane et al. and Wallace et al. utilized the MoM Health cohort participants;[33,41] however, Wallace et al. additionally included participants from a second cohort and, as a result, we retained both studies in our analysis. Lane et al. assessed sedentary behavior as time spent in a sitting or reclining position measured using an activity monitor.[33] Wallace et al. assessed time spent sitting through self-report on the Sedentary Behavior Two Domain Questionnaire (SB2D).[41] Regarding systolic blood pressure, only one study, Raguema et al., found a statistically significant positive correlation with sedentary behavior self-reported on the PPAQ (correlation coefficient = 0.25, $0.001 < P < 0.05$).[38] However, when the authors restricted their analysis to pre-eclampsia cases only, they found a negative and not statistically significant correlation (correlation coefficient = −0.05).

Regarding diastolic blood pressure, only Lane et al. found an association, specifically between high versus low sedentary behavior and diastolic blood pressure measured in the second trimester at a study visit (β [95% CI] = 4.9 [1.6–8.2]).[33] This association was not statistically significant with third-trimester blood pressure or blood pressure measured as part of a routine prenatal clinic visit. Similarly, none of the other studies found an association between sedentary behavior and diastolic blood pressure.

4.5 | Quality assessment

To summarize quality assessments, the following thresholds were used to convert Newcastle-Ottawa Scale assessments to Agency for Health Research and Quality (AHRQ) standards: good quality, 3 or 4 stars in selection domain AND 1 or 2 stars in comparability domain AND 2 or 3 stars in outcome/exposure domain; fair quality, 2 stars in selection domain AND 1 or 2 stars in comparability domain AND 2 or 3 stars in outcome/exposure domain; poor quality, 0 or 1 star in selection domain OR 0 stars in comparability domain OR 0 or 1 stars in outcome/exposure domain. Cross-sectional studies assessed using the AXIS tool were converted into poor-, fair-, and good-quality categories based on the proportion of 'yes' responses: poor quality, <12/20 (60%); fair quality, 12–15/20 (60%–80%); good quality >15/20 (80%). Detailed quality assessments for each extracted study result can be found in Appendix S3.

All extracted study results from the four case–control studies were determined to be of poor quality, most commonly due to a lack of comment regarding non-response rate, interviews not being blinded to case–control status, and inadequate control of important confounding variables based on pre-defined confounding domains (age and body mass index [BMI, calculated as weight in kilograms divided by the square of height in meters]).[35,38–40] Four of the five included cohort studies had results of good quality.[31–33,41] Results from Meander et al. were assigned poor to fair quality due to methodology involving participant recall of sedentary time during pregnancy (exposure) at 32–34 weeks gestation after the time of diagnosis of pre-eclampsia/gestational hypertension (outcome) which may have introduced potential for recall bias on the exposure depending on

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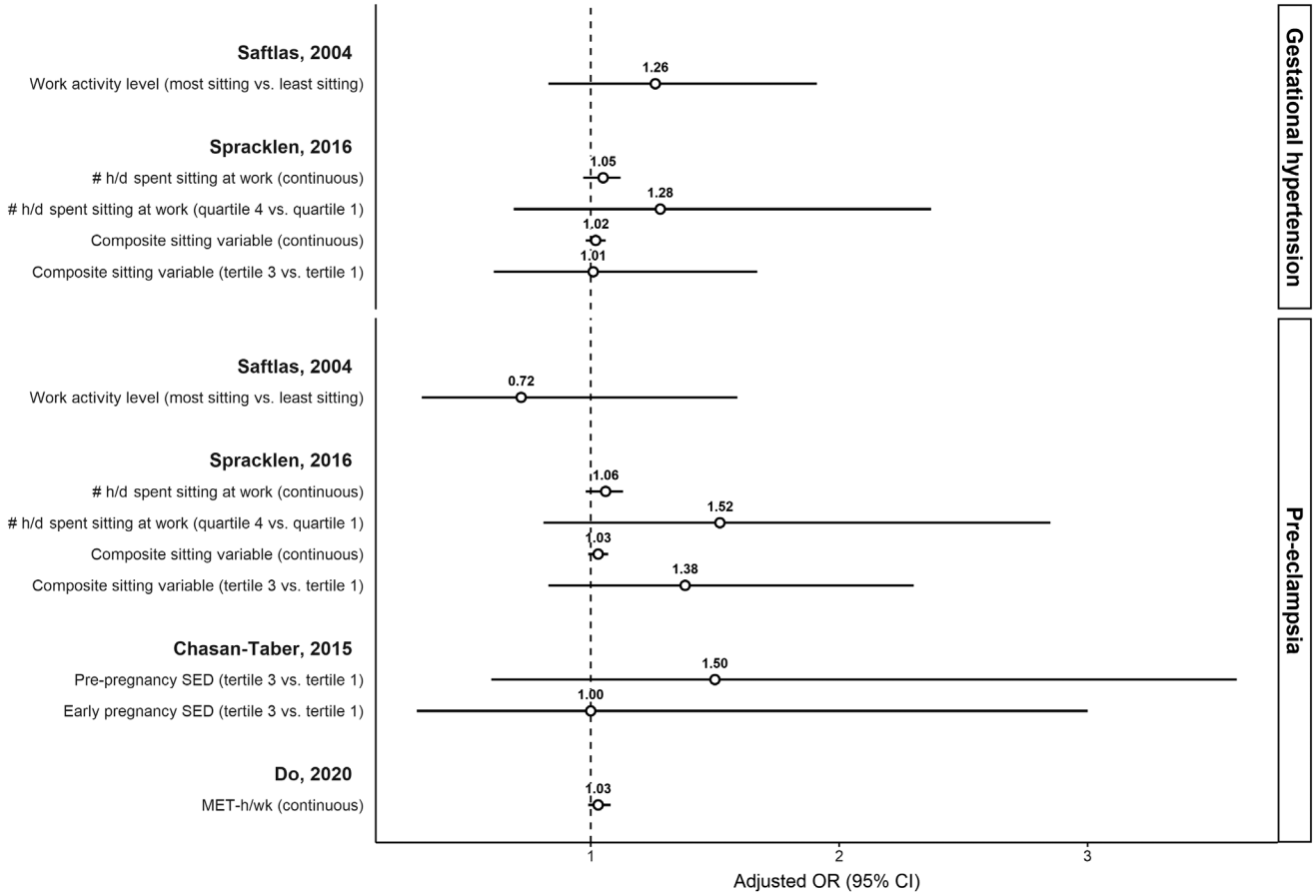

FIGURE 2 Reported adjusted odds ratios and 95% confidence intervals of studies assessing gestational hypertension and pre-eclampsia. CI, confidence interval; OR, odds ratio.

factors associated with the outcome.[37] Two of the three included cross-sectional studies were assigned good quality.[34,36] Results from Aburezq et al. were assigned fair quality due to a lack of sample size justification, self-reported blood pressure ascertainment, no description of non-responders, and no discussion of study limitations.[30] Additionally, participants were recruited postpartum, and therefore report of sedentary behavior during pregnancy is subject to recall bias and may be altered based on experiences during pregnancy, labor, or delivery.

5 | DISCUSSION

This systematic review based on 12 observational studies found limited evidence of an association between sedentary behavior and pre-eclampsia, gestational hypertension, total hypertensive disorders of pregnancy, or gestational blood pressure. The limited number of studies assessing this association as well as heterogeneity in the methods used to estimate sedentary behavior was a barrier to quantifying this relationship, indicating a need for continued investigation on this topic. Of the included studies, only two used an objective measure of sedentary behavior[33,36] and some of the exposure measures have not previously been validated. Although four studies found a statistically significant association between sedentary behavior and a hypertensive disorder of pregnancy in their most-adjusted results,[33,35,37,38] three of these studies were rated as either poor or fair quality using the Newcastle-Ottawa Scale.

We are aware of only one other systematic review that examined the relationship between sedentary behavior and gestational hypertensive disorders. In 2017, Fazzi et al. synthesized three studies on sedentary behavior during pregnancy and hypertensive disorders during gestation.[42] Two of these studies were included in our review, neither of which found a statistically significant association.[31,36] Fazzi et al. also included a study which found more gestational hypertension among women who worked in environments requiring sitting (e.g., sewing operators) as compared with women working in environments which were also sedentary but wherein they could move during working time (e.g., secretaries).[43] We excluded this study at the full-text stage as occupation was used as a marker for sedentary behavior and no quantitative estimate of sedentariness or time spent engaging in sedentary behavior was provided. Our findings build upon those by Fazzi et al. to suggest that there is little evidence of an association between sedentary behavior and pre-eclampsia or other hypertensive disorders of pregnancy.

There is a large body of evidence on the association between physical activity immediately prior to or during pregnancy and

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pre-eclampsia, demonstrating, for the most part, a protective effect of physical activity against pre-eclampsia.[15,44] Conversely, far less is known about sedentary behavior as a risk factor for pre-eclampsia. The scientific literature in this field has expanded only in recent years and only two of our included studies were published more than 10 years ago.[36,39] The disparity in results may be based on differences in the methods for estimating participants' sedentary behavior, as most included studies relied on self-reported behavior. In the case of highly sedentary individuals incorrectly classified as being less sedentary, the effect of sedentary behavior may have been underestimated in studies. Alternatively, decreasing sedentary behavior may be less important than increasing physical activity, in relation to reducing the risk of hypertensive disorders of pregnancy. Larger prospective studies are needed to better evaluate this association, and randomized trials evaluating the effect of decreasing sedentary behavior may also be useful.

The reliance on self-reported measures of sedentary behavior is a limitation in this field. Although there is no single best measure for capturing sedentary behavior and even objective measures such as accelerometers have problems, objective measures are considered more accurate and reliable relative to subjective measures.[45–47] Subjective measures may be affected by recall bias and social desirability bias,[45,48] which may substantially affect results. Conversely, objective measures cannot separate sedentary behavior according to domain (e.g., leisure, work, transportation) or type (e.g., television watching, driving, computer work). Future research should use objective measures of sedentary behavior when possible, either on its own or in combination with subjective measures. Additionally, the definition of sedentary behavior varied across studies, with broader definitions encompassing all activity below two METs and more limited definitions covering time spent sitting at work. We excluded many studies that also conceptualized sedentary behavior as low levels of physical activity, despite evidence that sedentary behavior and physical activity are independent risk factors for many disease outcomes.[24] This misclassification in the literature makes assessing relationships between sedentary behavior and outcomes more challenging.

Of the studies that conducted multivariable analysis, there was wide variety in the covariates included for adjustment. The variable most frequently adjusted for in analyses was BMI, but otherwise, confounder selection differed across studies. In terms of demographic and socioeconomic variables, age was adjusted for in six studies, and only two studies[33,36] adjusted for race and one[36] for an income variable (poverty-to-income ratio). Physical activity was taken into account in only one study.[39] Given that sedentary behavior is emerging as an independent risk factor from physical activity for many chronic diseases, adjustment for physical activity in future studies may be beneficial. None of the studies adjusted for comorbidities known to be risk factors for pre-eclampsia and other gestational disorders of pregnancy, such as chronic (non-gestational) hypertension and type 2 diabetes.

As with other systematic reviews, the present review has limitations due to the availability, quality, and heterogeneity of the identified primary studies. First, the methods and units for quantifying sedentary behavior varied across studies and many relied on self-reported sedentary behavior. Second, we identified few studies that met our inclusion criteria. Further research on this topic may provide further detail about the relationship of interest. Additionally, most studies included in the present systematic review were conducted in western counties. The generalizability of the findings may not apply to other settings with varying socioeconomic and lifestyle patterns. Finally, all included studies were observational as no experimental evidence was found. Future high-quality randomized trial evidence may provide a higher level of certainty regarding the relationship of interest.

6 | CONCLUSIONS

Overall, we found little evidence in the present systematic review to suggest a relationship between sedentary behavior and pre-eclampsia. Additional high-quality observational studies or randomized trials are warranted to reach more definitive conclusions.

AUTHOR CONTRIBUTIONS

LXP and SM contributed to the conception and design of the systematic review, registered the protocol in PROSPERO, conducted the literature search, extracted and interpreted the data, conducted quality assessment, and drafted the manuscript. All authors reviewed, revised, and approved the final manuscript.

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CONFLICT OF INTEREST STATEMENT

The authors have no conflicts of interest.

DATA AVAILABILITY STATEMENT

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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